1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 34

Transformations of Functions

34

Suppose $f(x) = x^2$

Sketch $y = f(x)$:

Then $f(x+2) =$

Sketch $y = f(x+2)$

What do you notice about the relationship between the graphs of $y = f(x)$ and $y = f(x+2)$?

Chapter 4 · Original slide 35

Transformations of Functions

35

This is all you need to remember when considering how transforming your function transforms your graph...

	Affects which axis?	What we expect or opposite?
Change **inside** $f()$		
Change **outside** $f()$		

Therefore...

$y = f(x-3)$

$y = f(x) + 4$

$y = f(5x)$

$y = 2f(x)$

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 36

Sketching transformed graphs

36

Sketch $y = x^2 + 3$

Sketch $y = \frac{2}{x+1}$

Chapter 4 · Original slide 37

More Examples

37

Sketch $y = x(x+2)$. On the same axes, sketch $y = (x-a)(x-a+2)$, where $a > 2$.

Sketch $y = x^2(x-4)$. On the same axes, sketch the graph with equation

$$y = (2x)^2(2x-4)$$

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 38

Reflections of Graphs

38

If $y = x(x+2)$, sketch $y = f(x)$ and $y = -f(x)$ on the same axes.

Chapter 4 · Original slide 39

Test Your Understanding

39

If $y = (x+1)(x-2)$, sketch $y = f(x)$ and $y = f(\frac{x}{3})$ on the same axes.

Sketch the graph of $y = \frac{2}{x} + 1$, ensuring you indicate any intercepts with the axes.

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 40

Exercise 4E – Pure Y1 page 74-75 (translations)

40

7 The point $P(4, -1)$ lies on the curve with equation $y = \mathrm{f}(x)$.
a State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x - 2)$.
b State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x) + 3$.

f
o

8 The graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \frac{1}{x}$ is translated so that the asymptotes are at $x = 4$ and $y = 0$. Write down the equation for the transformed function in the form $y = \frac{1}{x+a}$.

Chapter 4 · Original slide 41

Exercise 4E – Pure Y1 page 74-75 (translations)

41

11 a Sketch the graph of $y = x^3 + 4x^2 + 4x$. **b** The poin
coordinates $(-1, 0)$ lies on the curve with equation
$y = (x + a)^3 + 4(x + a)^2 + 4(x + a)$ where a is a constant. Find
the two possible values of a.

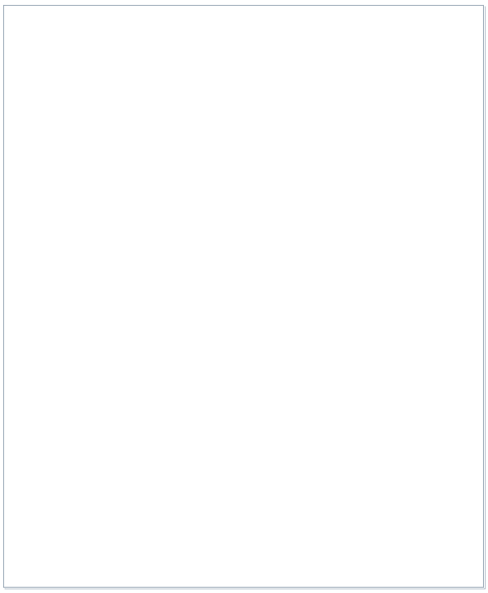
1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 42

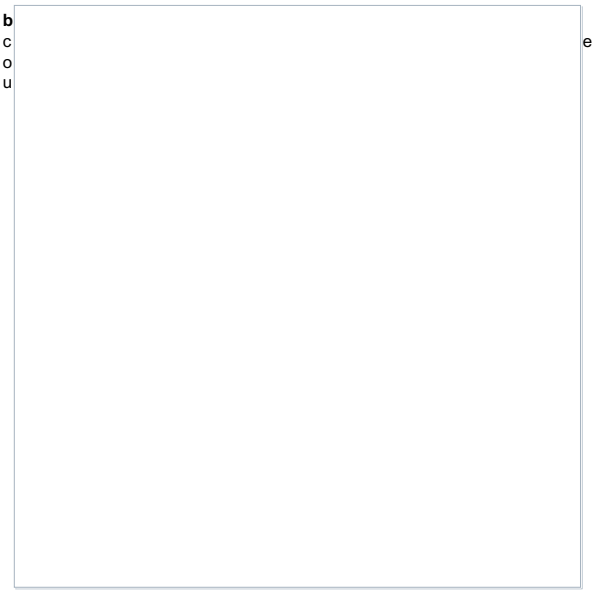
Exercise 4E – Pure Y1 page 74-75 (translations)

42

12 a Sketch the graph of $y = x(x+1)(x+3)^2$. **b** Find the possible values of b such that the point $(2, 0)$ lies on the curve with equation $y = (x+b)(x+b+1)(x+b+3)^2$.



b
c
o
u



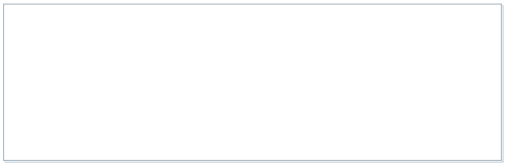
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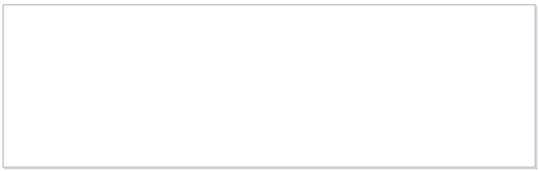
Chapter 4 · Original slide 43

Exercise 4F – Pure 1 Page 78 (stretches/reflections)

43

7 The point $P(2, -3)$ lies on the curve with equation $y = \mathrm{f}(x)$. **a** State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(2x)$. **b** State the coordinates that point P is transformed to on the curve with equation $y = 4\mathrm{f}(x)$.





8 The point $Q(-2, 8)$ lies on the curve with equation $y = \mathrm{f}(x)$. State the coordinates that point Q is transformed to on the curve with equation $y = \mathrm{f}(\frac{1}{2}x)$.



1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 44

Exercise 4F – Pure 1 Page 78 (stretches/reflections)

44

9 a Sketch the graph of $y = (x-2)(x-3)^2$. **b** The graph of $y = (ax-2)(ax-3)^2$ passes through the point $(1, 0)$. Find two possible values for a.

Chapter 4 · Original slide 45

Effect of transformation on specific points

45

Sometimes you will not be given the original function, but will be given a sketch with specific points and features you need to transform.
Where would each of these points end up?

$\mathbf{y=f(x)}$	$(4,3)$	$(1,0)$	$(6,-4)$
$y=f(x+1)$			
$y=f(2x)$			
$y=3f(x)$			
$y=f(x)-1$			
$y=f\left(\frac{x}{4}\right)$			
$y=f(-x)$			
$y=-f(x)$			

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 46

Test Your Understanding

46

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Q10

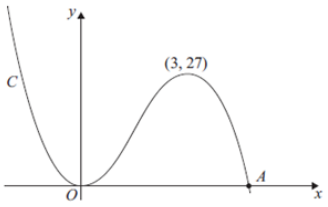


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = f(x)$, where $f(x) = x^2(9 - 2x)$. There is a minimum at the origin, a maximum at the point $(3, 27)$ and C cuts the x-axis at the point A.

(a) Write down the coordinates of the point A. (1)

(b) On separate diagrams sketch the curve with equation

(i) $y = f(x + 3)$,

(ii) $y = 2f(x)$.

On each sketch you should indicate clearly the coordinates of the maximum point and any points where the curves cross or meet the coordinate axes. (5)

The curve with equation $y = f(x) + k$, where k is a constant, has a maximum point at $(3, 10)$.

(c) Write down the value of k. (1)

(a)

(b)(i)

(ii)

(c)

Chapter 4 · Original slide 47

Exercise 4G Pure Y1 Pages 80

47

5 The point $P(2, 1)$ lies on the graph with equation $y = f(x)$.

a On the graph of $y = f(ax)$, the point P is mapped to the point $Q(4, 1)$. Determine the value of a.

b Write down the coordinates of the point to which P maps under each transformation

i $f(x - 4)$ **ii** $3f(x)$ **iii** $\frac{1}{2}f(x) - 4$

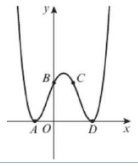
1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 48

Exercise 4G Pure Y1 Pages 80

48

6 The diagram shows a sketch of a curve with equation $y = \mathrm{f}(x)$. The points $A(-1, 0), B(0, 2), C(1, 2), D(2, 0)$ lie on the curve. Sketch the following graphs and give the coordinates of the points A, B, C, D after each transformation:



a $y + 2 = \mathrm{f}(x)$ b $\frac{1}{2}y = \mathrm{f}(x)$ c $y - 3 = \mathrm{f}(x)$ d $3y = \mathrm{f}(x)$ e $2y - 1 = \mathrm{f}(x)$